DDPM

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1 DDPMs

Concept

The idea behind Denoising Diffusion Probabilistic Models (or Diffusion Models for short) is to destroy the training data by adding noise, and then learn to reverse this process to generate new data.

Let us consider the MNIST dataset, which consists of images of handwritten digits from 0 to 9. There exists some manifold in the high-dimensional space of images, that contains all such images and the MNIST dataset must be a submanifold of it. Our aim is to sample new images from this manifold, but the problem is that we do not know the algebraic form of this manifold because it is too complex.

So we will take a point on the submanifold and add noise to it iteratively, to push it outside the submanifold. Then we will train a model to learn to remove the noise step by step, to get back into the manifold.

Forward Process

We start from a initial image x_0 and corrupt it as follows:

$$x_t = \sqrt{1 - \beta_t} x_{t-1} + \sqrt{\beta_t} \epsilon_t \tag{1}$$

Here, ϵ_t is sampled from an independent standard Gaussian distribution $\epsilon_t \sim \mathcal{N}(0, I)$ and the parameter $\beta_t \in (0, 1)$ controls the amount of noise added. The term $\sqrt{1 - \beta_t} x_{t-1}$ retains a fraction of the previous state.

The transition density of getting a particular x_t given some x_{t-1} is given by

$$q(x_t \mid x_{t-1}) = \mathcal{N}\left(x_t \mid \sqrt{1 - \beta_t} x_{t-1}, \beta_t I\right) \tag{2}$$

where $\mathcal{N}(x \mid \mu, \sigma^2)$ is the probability density at x.

The process is repeated for T steps. $\beta_1, ..., \beta_T$ is the variance schedule.

Forward Process (ii)

By reparameterizing

$$\begin{aligned}\alpha_t &= 1 - \beta_t \\ \overline{\alpha}_t &= \prod_{i=1}^{t} (1 - \beta_i)\end{aligned} \tag{3}$$

we can write

$$\begin{aligned}x_t &= \sqrt{\alpha_t} x_{t-1} + \sqrt{1 - \alpha_t} \epsilon_t \\ &= \sqrt{\alpha_t} \left(\sqrt{\alpha_{t-1}} x_{t-2} + \sqrt{1 - \alpha_{t-1}} \epsilon_{t-1} \right) + \sqrt{1 - \alpha_t} \epsilon_t \\ &= \sqrt{\alpha_t} \sqrt{\alpha_{t-1}} x_{t-2} + \sqrt{\alpha_t} \sqrt{1 - \alpha_{t-1}} \epsilon_{t-1} + \sqrt{1 - \alpha_t} \epsilon_t \\ &= \sqrt{\alpha_t} \sqrt{\alpha_{t-1}} x_{t-2} + \sqrt{1 - \alpha_t \alpha_{t-1}} \epsilon_{t,t-1} \quad \left(\because a\epsilon_1 + b\epsilon_2 = \sqrt{a^2 + b^2} \epsilon_3 \right) \\ &\vdots \\ x_t &= \sqrt{\overline{\alpha}_t} x_0 + \sqrt{1 - \overline{\alpha}_t} \overline{\epsilon}_t\end{aligned} \tag{4}$$

Forward Process (iii)

This way we can go to any arbitrary time step t for a given x_0. The transition probability becomes

$$q(x_t \mid x_0) = \mathcal{N}\left(x_t \mid \sqrt{\overline{\alpha}_t} x_0, (1 - \overline{\alpha}_t) I\right) \tag{5}$$

Since all α_i are in $(0, 1)$, for large enough T, $q(x_t \mid x_0) \to \mathcal{N}(x_T \mid 0, I)$, and so we get pure Gaussian noise.

Backward Process

Now we would like to find the probability density of the reverse process i.e. $q(x_{t-1} \mid x_t)$. More specifically, for training we want the model to learn to generate x_0, so we want to find $q(x_{t-1} \mid x_t, x_0)$.

By product rule, we can write

$$\begin{aligned} P(A, B, C) &= P(A \mid B, C)P(B \mid C)P(C) \\ &= P(B, A, C) = P(B \mid A, C)P(A \mid C)P(C) \\ \therefore P(A \mid B, C) &= \frac{P(B \mid A, C)P(A \mid C)}{P(B \mid C)} \end{aligned} \tag{6}$$

So

$$q(x_{t-1} \mid x_t, x_0) = \frac{q(x_t \mid x_{t-1}, x_0)q(x_{t-1} \mid x_0)}{q(x_t \mid x_0)} \tag{7}$$

Backward Process (ii)

Now

$$q(x_t \mid x_{t-1}, x_0) = q(x_t \mid x_{t-1}) \quad (\because \text{Markovian process})$$

$$\therefore q(x_{t-1} \mid x_t, x_0) = \frac{\mathcal{N}(x_t \mid \sqrt{\alpha_t} x_{t-1}, (1-\alpha_t)I)\mathcal{N}\left(x_{t-1} \mid \sqrt{\overline{\alpha}_{t-1}} x_0, (1-\overline{\alpha}_{t-1})I\right)}{\mathcal{N}\left(x_t \mid \sqrt{\overline{\alpha}_t} x_0, (1-\overline{\alpha}_t)I\right)} \tag{8}$$

For a multivariate isotropic Gaussian distribution, the probability density at $x \in \mathbb{R}^d$ is given by

$$\mathcal{N}(x \mid \mu, \sigma^2 I) = \frac{1}{\sigma(2\pi)^{\frac{1}{d}}} \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^T(x-\mu)\right) \tag{9}$$

Backward Process (iii)

Plugging this in and ignoring the coefficient, we get

$$q(x_{t-1} \mid x_t, x_0) = \exp\Bigg(-\frac{1}{2} \frac{\left(x_t - \sqrt{\alpha_t} x_{t-1}\right)^T \left(x_t - \sqrt{\alpha_t} x_{t-1}\right)}{1 - \alpha_t} \\ -\frac{1}{2} \frac{\left(x_{t-1} - \sqrt{\overline{\alpha}_{t-1}} x_0\right)^T \left(x_{t-1} - \sqrt{\overline{\alpha}_{t-1}} x_0\right)}{1 - \overline{\alpha}_{t-1}} \\ +\frac{1}{2} \frac{\left(x_t - \sqrt{\overline{\alpha}_t} x_0\right)^T \left(x_t - \sqrt{\overline{\alpha}_t} x_0\right)}{1 - \overline{\alpha}_t} \Bigg) \tag{10}$$

We would like to express this in the form of a Gaussian distribution $\mathcal{N}(x \mid \tilde{\mu}(x_t, x_0), \tilde{\sigma}^2(t) I)$ (so as to "subtract" the Gaussian noise we added earlier), so we need to find the mean and variance.

Backward Process (iv)

Looking at Equation 9, we can find $\tilde{\sigma}^2$ by finding the coefficient of $x_{t-1}^T x_{t-1}$, which is

$$\begin{aligned} &\frac{1}{2}\frac{\alpha_t}{1-\alpha_t} + \frac{1}{2}\frac{1}{1-\overline{\alpha}_{t-1}} \\ &= \frac{1}{2}\frac{1-\overline{\alpha}_t}{(1-\alpha_t)(1-\overline{\alpha}_{t-1})} \end{aligned} \tag{11}$$

$$\therefore \tilde{\sigma}^2(t) = \frac{(1-\alpha_t)(1-\overline{\alpha}_{t-1})}{1-\overline{\alpha}_t} \tag{12}$$

This has no dependence on x_i and hence can be written as

$$\tilde{\sigma}^2(t) = \tilde{\beta}(t) = \frac{1-\overline{\alpha}_{t-1}}{1-\overline{\alpha}_t}\beta_t \tag{13}$$

Going by the original paper [1], we assume this to be a constant rather than varying with t.

Backward Process (v)

Looking at Equation 9 again, we can find $\tilde{\mu}(x_t, x_0)$ by finding the vector v multiplied to x_{t-1}^T, which is

$$\frac{\tilde{\mu}(x_t, x_0)}{\tilde{\sigma}^2(t)} = v = \frac{\sqrt{\alpha_t} x_t}{1 - \alpha_t} + \frac{\sqrt{\overline{\alpha}_{t-1}} x_0}{1 - \overline{\alpha}_{t-1}} \tag{14}$$

$$\therefore \tilde{\mu}(x_t, x_0) = \frac{\sqrt{\alpha_t}(1 - \overline{\alpha}_{t-1}) x_t + \sqrt{\overline{\alpha}_{t-1}}(1 - \alpha_t) x_0}{1 - \overline{\alpha}_t} \tag{15}$$

This will only depend on x_t because we know $x_t(x_0, \overline{\epsilon}_t)$ (by Equation 4)

$$\therefore \tilde{\mu}(x_t) = \frac{1}{\sqrt{\overline{\alpha}_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \overline{\alpha}_t}} \overline{\epsilon}_t \right) \tag{16}$$

Therefore, to get $\mathcal{N}\left(x_{t-1} \mid \tilde{\mu}(x_t), \tilde{\beta}\right)$ we need to estimate the original noise that corrupted the image. We estimate this using a neural network and so we need to define a loss function for training.

Backward Process (vi)

If our estimated noise is $\epsilon_\theta(x_t, t)$, then the loss function is

$$L_t = E_{x_0,\epsilon_t}[\| \bar{\epsilon}_t - \epsilon_\theta(x_t, t) \|^2] \tag{17}$$

We train a UNet model as per [1]. It takes as input the noisy image x_t and the time step t, and outputs the estimated noise $\epsilon_\theta(x_t, t)$.

The derivations were based on [2].

2 Bibliography

[1] J. Ho, A. Jain, and P. Abbeel, "Denoising Diffusion Probabilistic Models," *arXiv preprint arxiv:2006.11239*, 2020.

[2] C. F. Higham, D. J. Higham, and P. Grindrod, "Diffusion Models for Generative Artificial Intelligence: An Introduction for Applied Mathematicians," 2023, doi: 10.48550/arxiv.2312.14977.